Special instructions for invigilators: None.

Information for Candidates:

Numbers in brackets in the right margin (e.g. [5]) indicate maximum marks for each section of each question.

The following constants may be used:

electron charge : $e = 1.6 \times 10^{-19}$ C

permittivity of free space : $\varepsilon_o = 8.85 \times 10^{-12}$ F/m

relative permittivity of silicon : $\varepsilon_r = 12$

Planck's constant : $h = 6.63 \times 10^{-34}$ J s

Boltzmann's constant : $k = 1.38 \times 10^{-23}$ J/K

speed of light : $c = 3.0 \times 10^{8}$ m/s

The eigenvalue equations for transverse electric (TE) modes in a symmetric slab waveguide of thickness d are:

$\kappa = k_{1x} tan(k_{1x} d/2)$ and $\kappa = -k_{1x}\, cot(k_{1x} d/2)$

Dispersion Parameter approximation: At 3 dBm dispersion power penalty, the dispersion parameter can be approximated as: $BLD\sigma_\lambda = 0.2$.

1. You should attempt all parts of this question. Short answers only are required; there is no need to re-state the questions in your answer book, but you should show any calculations you use to arrive at your answers, state any assumptions or approximations made, and give a brief (one or two lines) explanation where appropriate. All parts have equal value.

[20]

a) An optical signal arrives at the end of a 200 km length of fibre with 0.1% of the intensity launched at the transmitter end. Calculate the fibre loss coefficient in dB/km.

Solution:

The loss is: $10\log\frac{I_2}{I_1} = -30\ dB$

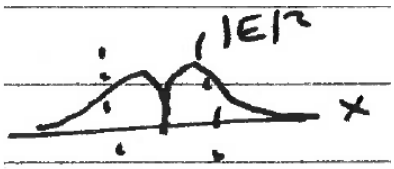
The loss is linear in the logarithmic scale so per unit length:

$$-\frac{30dB}{200km} = -0.15\ dB/km$$

b) A certain slab waveguide supports two TE modes, *m=0* and *m=1*. For the *m=1* mode, is the optical intensity higher at the centre of the guide, or at the core-cladding boundary? Give a brief explanation.

Solution:

Odd numbered modes are anti-symmetric. They have zero intensity at the center and therefore the intensity is higher at the boundary.



c) A certain slab waveguide supports 20 TE modes at a nominal wavelength $\lambda_o = 720$ nm. For an increased wavelength of $\lambda_o = 1300$ nm, would you expect the guide to support fewer, more or the same number of TE modes? Give a brief explanation.

Solution:

The number of supported modes is reduced as the wavelength is increased. Given the large number of supported modes and the large wavelength increase, there will certainly be fewer modes supported at the new higher wavelength.

d) If at some instant in time a 500 m length of optical fibre contains 7500 bits of propagating information, what is the bit rate of this signal?

Solution:

Estimating a refractive index of $n = 1.5$, the group velocity will be:

$$v_g = \frac{c}{n} = 2 \cdot 10^8 \text{m/s}$$

The time that corresponds to a 500 m length of travel will be: $\tau = \frac{L}{v_g} = \frac{500}{2} \cdot 10^{-8} = 2.5\ \mu\text{s}$

Therefore, the bitrate will be: $B = \frac{7500\text{ bit}}{2.5\ \mu\text{s}} = 3\ \text{Gbps}$

e) Why is silicon not a suitable material for constructing detectors for optical systems operating at nominal wavelength 1650 nm?

Solution:

Silicon has a bandgap energy that corresponds to a wavelength of 1.11 μm (1.12 eV). Photons of higher wavelength have lower energy and can't be absorbed for electron-hole pair generation. Therefore, Si can't operate as a detector at 1650 nm.

f) Give a practical disadvantage of polarisation maintaining fibre in comparison to conventional single mode fibre.

Solution:

The alignment of two fibres is more difficult because their rotation as well as position need to be matched.

g) Silica fibres generally have higher attenuation at 1.3 μm than at 1.5 μm wavelength. Which attenuation mechanism is primarily responsible for this higher attenuation and why?

Solution:

Rayleigh scattering, as it dominates fibre attenuation at short wavelengths. The attenuation coefficient of Rayleigh scattering is proportional to λ^{-4} and therefore it is much higher at smaller wavelengths.

h) A laser diode with quantum efficiency 0.80 has a slope efficiency of 1.16 W/A. Calculate the output (free space) wavelength.

Solution:

$$S = \frac{\eta h c}{e\lambda}, \; hence \; \lambda = \frac{\eta c h}{eS} = 0.88 \; \mu m$$

i) Calculate the optimal thickness and refractive index of an anti-reflection coating acting between a glass of index 1.48 and air, for a free space wavelength of 0.75 μm.

Solution:

$$n = \sqrt{1.52 \cdot 1} = 1.22$$

$$\tau = \frac{\lambda}{4n} = 0.154 \; \mu m$$

j) Briefly explain why an avalanche photodiode usually has the avalanche region confined to the base of the depleted region.

Solution:

In this way very few photons are absorbed in the avalanche region, so the region receives all the photo-generated electrons, and none of the holes. This makes the avalanche factor more predictable and controllable.

2. A symmetric slab waveguide has core and cladding indices of n_1 = 1.46 and n_2 = 1.45 respectively. The guide is shown in Fig. 2.1. The core thickness d is 12 μm. A certain mode, at free-space wavelength 1.50 μm has an effective index of n' = 1.45264.

a) Define the boundary conditions required at the core-cladding interfaces for TE modes, with respect to the electric field distribution $E(x)$. Show that the effective index given above is consistent with these conditions.

[10]

b) Find the mode number m and sketch the corresponding field profile $E(x)$ for the mode having this value of n'.

[5]

c) What is the total number of TE modes supported by the guide at this wavelength?

[5]

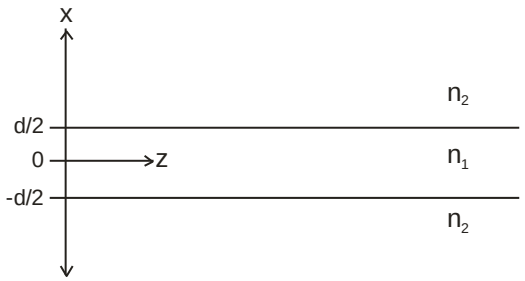


Figure 2.1 Slab waveguide

Solution:

a) Let $E_1(x)$ and $E_2(x)$ be the field in the core ($-\frac{d}{2} < x < \frac{d}{2}$) and in the cladding ($x \geq \frac{d}{2}$) respectively. Then, the boundary conditions are:

$$E_1\left(\frac{d}{2}\right) = E_2\left(\frac{d}{2}\right)$$

$$\frac{dE_1}{dx}\left(\frac{d}{2}\right) = \frac{dE_2}{dx}\left(\frac{d}{2}\right)$$

For even modes:

$$E_1(x) = A\cos k_{1x}x$$

$$E_2(x) = Be^{-\kappa x}$$

$$\frac{dE_1(x)}{dx} = -Ak_{1,x}\sin k_{1,x}x$$

$$\frac{dE_2(x)}{dx} = -B\kappa e^{-\kappa x}$$

Dividing the boundary conditions:

$$k_{1,x}\tan\left(\frac{k_{1,x}d}{2}\right) = \kappa$$

For odd modes:

$$E_1(x) = Asin(-k_{1,x}x)$$

$$\frac{dE_1(x)}{dx} = -Ak_{1,x}\cos(-k_{1,x}x)$$

Dividing as before gives:

$$-k_{1,x}\cot\left(\frac{k_{1,x}d}{2}\right) = \kappa$$

Additionally:

$$k_{1,z} = k_{2,z} = \beta$$

The consistency of the given values with the above conditions can be checked as follows:

$$\beta = n'k_0 = 1.45264 \cdot \frac{2\pi}{1.5} = 6.0848\ \mu m^{-1}$$

Then κ can be calculated as:

$$\kappa = \sqrt{\beta^2 - n_2^2k_0^2} = 0.3667\ \mu m^{-1}$$

And $k_{1,x}$ as:

$$k_{1,x} = \sqrt{n_1^2k_0^2 - \beta^2} = 0.6133\ \mu m^{-1}$$

Checking the condition for even m:

$$\kappa = k_{1,x}\tan\left(\frac{k_{1,x}d}{2}\right) = 0.3661\ \mu m^{-1}$$

This value is close enough to the value of κ calculated above.

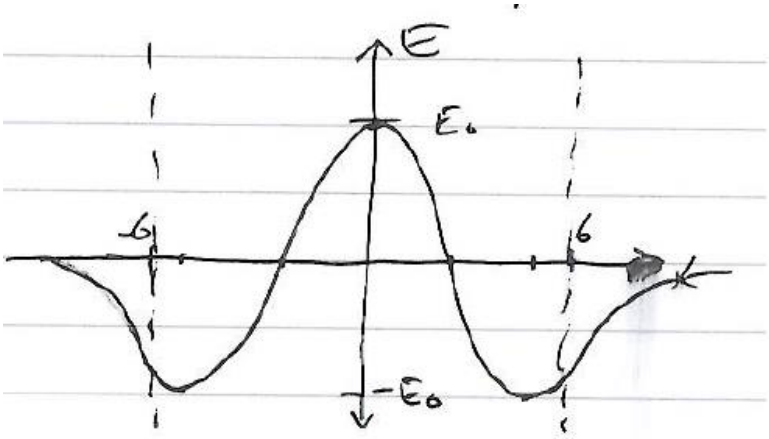
b)

The mode can be checked from the $\frac{k_{1x}d}{2}$ value which is 3.68, hence between π and $\frac{3\pi}{2}$.

Therefore, $m = 3$. The field profile can be sketched by calculating the axis-crossing points within $-\frac{d}{2} \le x \le d/2$:

$$E_1(0) = A$$

$$E_1(x) = 0 \Leftrightarrow cosk_{1,x}x = 0 \Leftrightarrow x = \pm 2.56\ \mu m$$



c) The condition for mode m support is:

$$NA \cdot k_0 \cdot d \geq m \cdot \pi$$

$$m \leq 2.73$$

Therefore the highest mode supported is $m = 2$. Three modes in total!

3. a) Write the equations that define the phase and group velocity of a propagating electromagnetic wave. What is the main physical significance of phase velocity?

[6]

b) At a given spectrum band, the refractive index of a sample of silica glass can be approximated by

$$n(\lambda_o) \cong D_o + D_1 \lambda_o^{-2} - D_2 \lambda_o^{2} \qquad (5.1)$$

where $D_o = 1.41$, $D_1 = 0.002$ μm^2 and $D_2 = 0.003$ μm^{-2}.

i) Calculate the phase and group velocities v_p and v_g in this material, as functions of D_o, D_1, D_2, and λ_o and the speed of light c. Which of the two is smaller at this spectrum band?

[6]

ii) Derive an expression for the phase change $\Delta\varphi$ experienced by a wave of nominal wavelength λ_o propagating a distance L.

[4]

iii) Derive an expression of wavelength at which dispersion is zero, as a function of D_1 and D_2. Calculate the wavelength for the values given above.

[4]

Solution:

a) The phase velocity is defined as: $v_p = \frac{\omega}{k}$

The group velocity is defined as: $v_g = \frac{d\omega}{dk}$

The phase velocity describes the speed at which each specific phase of a sinusoidal wave travels through space.

b)

i) For the phase velocity: $v_p = \frac{c}{n} = \frac{c}{D_0 + D_1\lambda_0^{-2} - D_2\lambda_0^2}$

For the group velocity: $v_g = \frac{1}{\frac{dk}{d\omega}}$

$$k = nk_0 \qquad and \qquad \omega = ck_0$$

$$\frac{dk}{d\omega} = \frac{dn}{d\omega}k_0 + \frac{n}{c}$$

Hence:

$$\frac{1}{v_g} = \frac{2\pi}{\lambda_0}\frac{dn}{d\lambda_0}\frac{d\lambda_0}{d\omega} + \frac{n}{c}$$

Now:

$$\frac{d\lambda_0}{d\omega} = -\frac{\lambda_0}{\omega} = -\frac{\lambda_0}{ck_0}$$

Thereby:

$$\frac{1}{v_g} = \frac{1}{c}(n - \lambda_0\frac{dn}{d\lambda_0})$$

But:

$$\frac{dn}{d\lambda_0} = -2D_1\lambda_0^{-3} - 2D_2\lambda_0$$

So finally:

$$v_g = \frac{c}{D_0 + 3D_1\lambda_0^{-2} + D_2\lambda_0^2}$$

The denominator of v_g is larger than that of v_p and therefore $v_p > v_g$.

ii) The phase change will be:

$$\Delta\varphi = 2\pi \cdot \frac{L}{\lambda} = \frac{2\pi nL}{\lambda_0} = 2\pi L(D_0\lambda_0^{-1} + D_1\lambda_0^{-3} - D_2\lambda_0)$$

iii) Zero dispersion is at:

$$\frac{d^2n}{d\lambda_0^2} = 0$$

$$\frac{d}{d\lambda_0}(-2D_1\lambda_0^{-3} - 2D_2\lambda_0) = 0$$

$$6D_1\lambda_0^{-4} - 2D_2 = 0$$

Hence:

$$\lambda_0 = \sqrt[4]{\frac{3D_1}{D_2}} = 1.189\ \mu m$$

4. a) Describe and discuss the important attenuation mechanisms in optical fibres, and the influence these have in choice of operating wavelengths. Use diagrams and equations where appropriate.

[6]

b) An optical receiver can be implemented as a photodiode in series with a resistor, followed by a voltage amplifier. State the principal noise sources in this case, and draw a noise equivalent circuit for the receiver.

[4]

c) A certain optical link has a receiver with noise equivalent power (NEP) of 9.0 pW/√Hz, at the end of 100 km of fibre having attenuation of 0.4 dB/km at the operating wavelength of 1210 nm. The transmitted power is 5 mW and the photodetector quantum efficiency can be approximated as 1.

i) Determine whether shot noise or receiver noise is greater in this case;

[4]

ii) Considering only the dominant noise source as determined in (i), find the maximum bit rate for an optical *SNR* of 12;

[4]

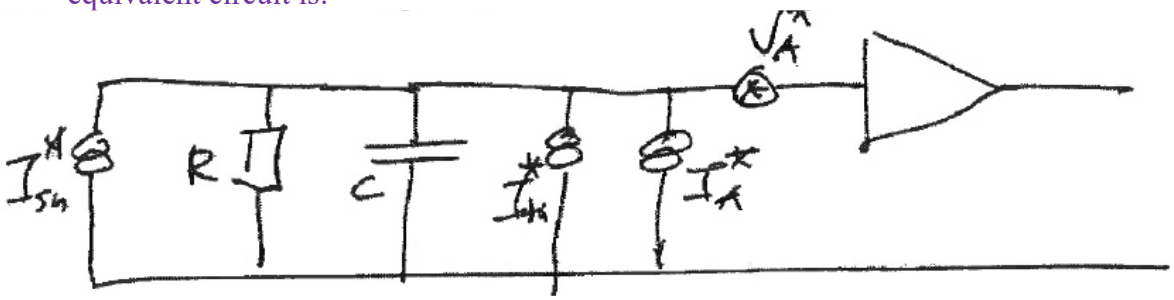
iii) For the bit rate determined in (ii), find an approximate value for the maximum dispersion coefficient for this fibre above which the dispersion penalty would exceed 3 dB. Assume a transmitter spectral width σ_λ of 1.2 nm.

[2]

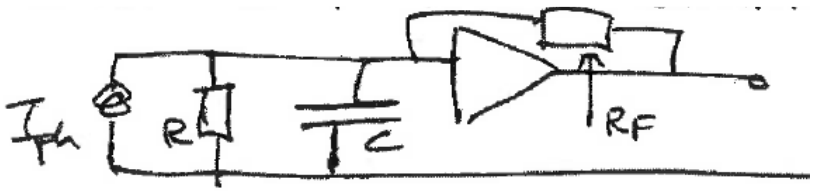
Solution:

a) The discussion should include absorption at UV and IR as well as Rayleigh scattering and the underlying mechanisms. Also, O-H bond absorption due to water remaining in the fibre should be mentioned. A plot of these should be sketched over a sensible wavelength range. Bending loss should be mentioned. The λ – dependence of Rayleigh scattering ($\propto \lambda^{-4}$) should be indicated.

b) Noise sources are shot noise, thermal noise and amplifier voltage and current noise. The equivalent circuit is:



An alternative configuration is the transimpedance amplifier. In this case the photodiode is effectively grounded, so the frequency dependence of gain due to R and C is eliminated:



This also means that the frequency dependence of *SNR* is largely eliminated.

c)i) To compare the noise sources, they must be expressed in terms of photocurrent:

$$SNR_{opt} = \frac{I_{ph}}{\left(\Sigma I^{*2}\right)^{1/2} \Delta f^{1/2}}$$

For shot noise: $I_{sh}^{*\,2} = 2eI_{ph}$

For the receiver noise, the *SNR* can be expressed as:

$$SNR_{opt} = \frac{\Phi_R}{NEP\Delta f^{\frac{1}{2}}} = \frac{R\Phi_R}{RNEP\Delta f^{\frac{1}{2}}} = \frac{I_{ph}}{[R^2NEP^2]^{\frac{1}{2}}\Delta f^{\frac{1}{2}}}$$

So we need to compare: $I_R^{*\,2} = R^2NEP^2$ with I_{sh}^{*}.

$$R = \frac{\eta e\lambda}{hc} = 0.973 \text{ A/W}$$

$$I_R^{*\,2} = 7.674 \cdot 10^{-23} \text{A}^2/\text{Hz}$$

For shot noise:

$$\Phi_R = \Phi_T - loss$$

$$5\ mW = 6.99 \text{ dBm}$$

$$\Phi_R = 6.99\ dBm - 0.4 \cdot 100 = -33 \text{ dBm}$$

$$\Phi_R = 0.5 \text{ µW}$$

$$I_{ph} = R\Phi_R = 0.49 \text{ µA}$$

$$I_{sh}^{*\,2} = 2eI_{ph} = 1.56 \cdot 10^{-25} \text{A}^2/\text{Hz}$$

The shot noise is 2 orders of magnitude smaller, so the receiver noise dominates.

ii) $$\Delta f^{1/2} = \frac{\Phi_R}{SNR_{opt} \cdot NEP} = 5.05 \cdot 10^3 \text{ Hz}^{1/2}$$

$$B = 2\Delta f = 51 \text{ Mbps}$$

iii) At 3 dBm dispersion power penalty, the dispersion parameter can be approximated as:

$$BLD\sigma_\lambda = 0.2$$

From this value:

$$D_{max} = \frac{0.2}{10^5 \cdot 51 \cdot 10^6 \cdot 1.2 \cdot 10^{-9}} = 32.7 \cdot 10^{-6} \frac{\text{s}}{\text{m}^2} = 32.7 \frac{\text{ps}}{\text{nm} \cdot \text{km}}$$

5. a) A silicon p-n photodiode (Fig. 5.1) has a depletion layer thickness of w, and p and n doping levels respectively of N_A and N_D, with $N_A = 5N_D = 4\times10^{20}$ m^{-3}. The quantities w_p and w_n are the depleted widths in the p and n regions respectively. A reverse bias voltage V_b is applied. Find an expression for the full depletion width w as a function of V_b, and the value of V_b for which w =5 μm.

[8]

b) Neglecting Fresnel reflection, find an expression for the quantum efficiency η of the photodiode of (a) if the total p region thickness h = 8 μm, and the absorption coefficient $\alpha = 0.866\times10^5$ m^{-1}. Hence find the value of V_b for which $\eta = 0.80$.

[6]

c) Give a typical approximate value for the saturation drift velocity of electrons in a semiconductor. For the photodiode of part (a) and an electron mobility in the silicon of 0.12 m^2/Vs, find the bias voltage for which the electron drift velocity is above this saturation value in half of the depleted region.

[6]

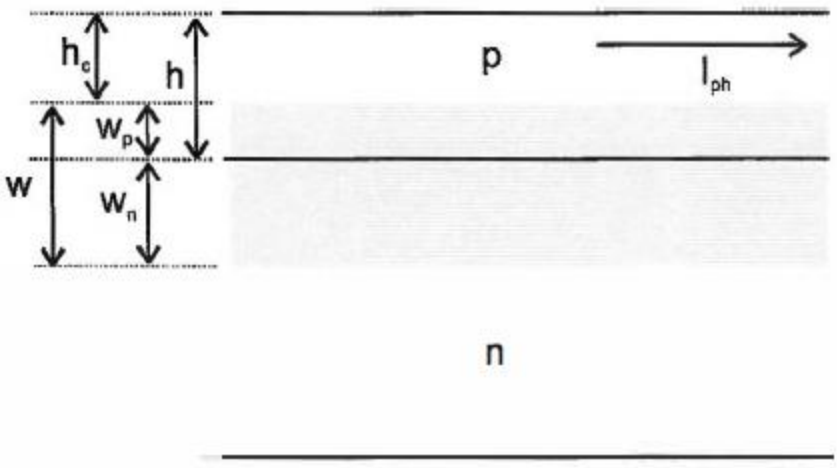


Figure 5.1: p-n photodiode.

Solution:

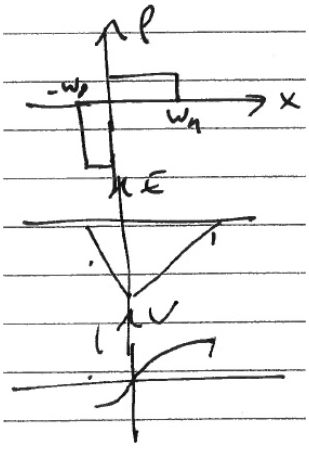
a)
Charge conservation:

$$w_n N_D = w_p N_A$$

Hence: $w_n = 5w_p$ and $w = 6w_p$

$$E_{MAX} = \frac{w_p N_A e}{\varepsilon}$$

$$V_b = \frac{1}{2} E_{MAX} w = \frac{w^2 N_A e}{12\varepsilon}$$



$$w = \sqrt{\frac{12\varepsilon(V_b + V_0)}{eN_A}}$$

For $w = 5$ µm*:* $V_b = 1.255$ V

b)

$$\eta = e^{-a(h-wp)} - e^{-a(h+w_n)} = e^{-ah}(e^{aw_p} - e^{-aw_n}) = e^{-ah}(e^{\frac{aw}{6}} - e^{-\frac{5aw}{6}})$$

$e^{-ah} = 0.500$

For, $\eta = 0.80$*,* setting $z = aw/6$*:*

$(e^z - e^{5z}) = 1.6$

Solving numerically gives z = 0.52, hence: w = 36 µm.

Then:

$$V_b = 65.16 \text{ V}$$

c)

A typical saturation value is $v_{SAT} = 10^5$ m/s

The corresponding field is:

$$E_{sat} = \frac{v_{sat}}{\mu_e} = 8.33 \cdot 10^5 \text{ V/m}$$

In order for half the depletion region to be in a field above this value:

$$E_{MAX} \geq 2E_{sat} = 1.67 \cdot 10^6 \text{ V/m}$$

Then:

$$w = E_{sat} \cdot \frac{\varepsilon}{e} \cdot \frac{(N_D + NA)}{N_D \cdot N_A} = 16.6 \text{ µm}$$

And:

$V_b = 13.83$ V